

Retrieval-Augmented Generation

Retrieval-Augmented Generation, or RAG, combines information retrieval with language generation. Instead of answering only from knowledge learned during training, a system searches an external collection for evidence relevant to a user's question. The collection may contain manuals, policies, reports, or other reference documents.

The basic workflow is query -> retrieval -> context -> generation. First, the user submits a query. Next, the retriever finds relevant passages. The application places selected passages into the model's context alongside the question and instructions. Finally, the model generates an answer using that context.

For example, a question about a company's vacation policy can retrieve the employee handbook. The answer can then describe the policy and cite the supporting passage. Updating the collection makes new information available without retraining the language model.

Grounding and Incorrect Answers

Grounding means connecting an answer to reliable evidence supplied in the context. A grounded response reflects what the retrieved passages actually support and identifies its sources when possible. Hallucination occurs when a model produces unsupported or invented claims, even when the wording sounds confident and convincing.

RAG can reduce hallucination, but it does not guarantee correctness. Retrieval may miss the relevant passage or return an outdated policy. Different sources may conflict. Important qualifications may be absent from the selected context, and the model may misread evidence or combine unrelated facts.

A useful system checks whether the available evidence is sufficient before answering. It should acknowledge missing information, preserve relevant conditions, and avoid treating unsupported assumptions as facts. Evaluating both retrieved evidence and generated answers helps reveal where failures occur.